Chao-Wei Huang

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Google Scholar
Homepage

EDUCATION

National Taiwan University 2018/09 - present
Ph.D. , Computer Science and Information Engineering (CSIE) (expected Aug. 2024)
Advisor: Yun-Nung (Vivian) Chen

National Taiwan University 2014/09 - 2018/06
B. S., Computer Science and Information Engineering (CSIE) GPA: 3.96/4.3

RESEARCH INTERESTS

- Natural Language Processing (NLP)
- Dialogue Systems
- Machine Translation
- Open-domain Retrieval

SELECTED PUBLICATIONS

[1] Chao-Wei Huang, Hui Lu, Hongyu Gong, Hirofumi Inaguma, Ilia Kulikov, Ruslan Mavlyutov, Sravya Popuri. Investigating Decoder-only Large Language Models for Speech-to-text Translation. In *Proc. of INTERSPEECH 2024* (**Interspeech 2024**), Kos Island, Greece, September 1-5, 2024. [Paper]

[2] Chao-Wei Huang, Tsu-Yuan Hsu, Chen-An Li, Chen-Yu Hsu, Yun-Nung Chen. Unsupervised Multilingual Dense Retrieval via Generative Pseudo Labeling. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (**EACL 2024**)*, Malta, March 17-22, 2024. [Code] [Paper]

[3] Chao-Wei Huang, Chen-Yu Hsu, Tsu-Yuan Hsu, Chen-An Li, Yun-Nung Chen. CONVERSER: Few-Shot Conversational Dense Retrieval with Synthetic Data Generation. In *Proceedings of the 24th Annual Meeting of the Special Interest Group on Discourse and Dialogue (**SIGDIAL 2023**)*, Prague, Czechia, September 11-15, 2023. [Code] [Paper]

[4] Tsu-Yuan Hsu*, Chen-An Li*, Chao-Wei Huang, and Yun-Nung Chen. Visually-Enhanced Phrase Understanding. In *Proceedings of The 61st Annual Meeting of the Association for Computational Linguistics (**ACL 2023**)*, Toronto, Canada, July 9-14, 2023. [Paper]

[5] Chao-Wei Huang*, Shang-Chi Tsai*, and Yun-Nung Chen. Modeling Diagnostic Label Correlation for Automatic ICD Coding. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (**NAACL 2021**)*, Online, June 6-11, 2021. [Code] [Paper]

[6] Chao-Wei Huang and Yun-Nung Chen. Learning Spoken Language Representations with Neural Lattice Language Modeling. In *Proceedings of The 58th Annual Meeting of the Association for Computational Linguistics (**ACL 2020**)*, Seattle, USA, July 5-10, 2020. [Code] [Paper] [Slides]

[7] Shang-Yu Su, Chao-Wei Huang, and Yun-Nung Chen. Towards Unsupervised Language Understanding and Generation by Joint Dual Learning. In *Proceedings of The 58th Annual Meeting of the Association for Computational Linguistics (**ACL 2020**)*, Seattle, USA, July 5-10, 2020. [Code] [Paper]

[8] Chao-Wei Huang and Yun-Nung Chen. Learning ASR-Robust Contextualized Embeddings for Spoken Language Understanding. In *Proceedings of 2020 IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP 2020**)*, Barcelona, Spain, 2020. [Code] [Paper] [Slides] [Presentation]

[9] Chao-Wei Huang and Yun-Nung Chen. Adapting Pretrained Transformer to Lattices for Spoken Language Understanding. In *Proceedings of 2019 IEEE Workshop on Automatic Speech Recognition and Understanding (**ASRU 2019**)*, Sentosa, Singapore, December 14-18, 2019. [Code] [Paper]

WORKING EXPERIENCES

Meta

Research Scientist Intern, FAIR 2023/07 - 2023/11

Research on leveraging large language models to end-to-end speech translation. We examined architectural designs and fine-tuning techniques for improved speech-to-text translation.

Research Intern (AI) 2022/09 - 2023/01

Research on evaluating and improving translation accuracy of named entities. We found that nearest neighbor methods improve named entity translation quality.

Amazon

Applied Science Intern, AlexaAI 2021/07 - 2021/10

Research on transfer learning of dialogue state tracking with locale differences and ASR robustness.

Applied Science Intern, AlexaAI 2020/03 - 2020/06

Organization of DSTC9 track 1, focusing on knowledge selection and knowledge-grounded response generation in dialogue systems.

Taiwan AI Labs 2017/07 - 2019/12

Software Engineering Intern, Speech Recognition and Synthesis Group

Developing ASR system that serves as backbone of several products (e.g., Yating Transcriber, Yating Input Method, Yating Pianist). The ASR-assisted products have over a hundred thousand installations.

- Acoustic model data preparation, training and optimization.
- Language model data collection, training and optimization.

National Taiwan University

Teaching Assistant

CSIE5431 Applied Deep Learning (2019 Spring, 2020 Spring, 2021 Spring, 2022 Spring, 2023 Fall)

HONORS & AWARDS

1st Place, Formosa Grand Challenge - Talking to AI 2019/03

The Formosa Grand Challenge(FGC) is a challenge held by the Ministry of Science and Technology. The participants are required to develop a spoken question answering system that is capable of answering questions in spoken content. We won the 1st place overall among over 100 teams.

TECHNICAL SKILLS

Programming Languages

Python, C++

Deep Learning Frameworks

PyTorch, TensorFlow, Keras

LANGUAGES

Mandarin (native)

English (proficient)